

KIARASH VAZIRI GOODARZI

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RESEARCH INTERESTS

- Computational Neuroscience
- Self-supervised Representation Learning
- Applied Deep Learning
- Time-series Modeling

EDUCATION

University of Southern California Aug. 2023 - Present
Doctor of Philosophy, **Electrical and Computer Engineering**
Viterbi School of Engineering
Advisor: Prof. Maryam Shanechi

University of Southern California Aug. 2023 - May 2026
Master of Science, **Electrical and Computer Engineering**
Viterbi School of Engineering
GPA: **3.83/4**

Sharif University of Technology Sep. 2019 - July 2023
Bachelor of Science, **Electrical Engineering**
GPA: **17.80/20**

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Southern California August 2023 - Present

- Develop nonlinear state-space models for forecasting and representation learning from multisite human intracranial EEG.
- Develop predictive models of deep brain stimulation effects on intracranial neural dynamics.
- Design and conduct intracranial EEG and scalp EEG experiments in hospital and laboratory settings.

Research Intern, University of British Columbia March - October 2022

- Analyzed physiological signals—including PPG, EDA, and heart rate—from wearable recordings of patients with Parkinson’s disease to study non-invasive neuromodulation.
- Applied signal processing and system-identification methods, developed autoencoder-based anomaly detection, and built ML models for blood-pressure prediction.

Research Intern, Biosen Group May - August 2021

- Designed and developed an ECG patch for cardiac monitoring and atrial fibrillation detection using feature-engineered ML classifiers.

B.Sc. Thesis, Sharif University of Technology 2023

- Developed transfer-learning methods using graph kernels for cross-subject EEG classification.

TECHNICAL SKILLS

Programming & Computing	Python, MATLAB, CUDA, Linux, Git, SLURM, HPC
Deep Learning	PyTorch, transformers, state-space models, generative modeling
Scientific Computing	Signal processing, SciPy, NumPy, scikit-learn

PUBLICATIONS & CONFERENCES

- **Vaziri K.**, Loganesian L.L., Jo H.C., Vera R., Liu C., Lee B., Shanechi M.M., “Nonlinear Dynamical Modeling of Human Intracranial Brain Activity with Flexible Inference,” arXiv preprint arXiv:2512.22785, December 2025.
- **Vaziri K.**, Loganesian L.L., Jo H.C., Vera R., Liu C., Lee B., Shanechi M.M., “Nonlinear Flexible Modeling of Multi-site, Multi-subject Human Intracranial Neural Activity,” IEEE EMBS 12th International Conference on Neural Engineering, San Diego, November 2025.
- **Vaziri K.**, Loganesian L.L., Jo H.C., Vera R., Liu C., Lee B., Shanechi M.M., “Nonlinear Flexible Modeling of Multi-site Human Intracranial Neural Activity,” Annual Meeting, Symposium for Neuroscience (SfN), San Diego, November 2025.
- Vatankhah M*, **Vaziri K***, Doughan M, Copello F, Younessian F. Automated Detection of Clear-Aligner Attachments from STL Dental Scans: A Deep-Learning Tool for Attachment Segmentation and Tooth-Surface Reconstruction. In preparation for *Orthodontics and Craniofacial Research*. (* Co-first author)
- Baraeinejad B., Shayan M.F., Vazife A.R., Rashidi D., Hamedani M.S., Tavolinejad H., Razmara P., Gourji P., **Vaziri K.**, Vashaei D., Fakharzadeh M., “Design and Implementation of an Ultralow-Power ECG Patch and Smart Cloud-Based Platform,” *IEEE Transactions on Instrumentation and Measurement*, April 2022.

RELEVANT COURSEWORK

University of Southern California: Reinforcement Learning and Large Language Models, Advanced Natural Language Processing, Deep Learning and Its Application, Inference and Estimation: Theory and Algorithm, Random Processes, Data Analysis and Control Techniques for Neurotechnology, Probability in Electrical and Computer Engineering

Sharif University of Technology: Deep Learning, Convex Optimization, Introduction to Machine Learning, Linear Algebra, Computational Intelligence, Digital Signal Processing, Algorithm & Data Structures, Probability and Statistics in Engineering, Object Oriented Programming

AWARDS

- **Annenberg Graduate Fellowship Research and Creative Project Symposium Award**, University of Southern California, 2024
- **Annenberg Fellowship**, Ming Hsieh Department of Electrical and Computer Engineering, University of Southern California, 2023.
- Undergraduate High Honour Student, Sharif University of Technology
- **Ranked in the top 0.1% among 160,000 students** taking part in the National University Entrance Exam in the Mathematics branch, Iran, 2019.